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S11 – 07

Aim: Write a program for Merge sort using Object oriented language (Java, C++) and Haskell.

Java Code:

```
import java.util.*;

class MergeSort {

   public static void merge(int[] array, int[] new_array_1, int[] new_array_2, int left, int right)
{

      int i = 0, j = 0, k = 0;

      while (i < left && j < right) {

         if (new_array_1[i] <= new_array_2[j]) {

            array[k++] = new_array_1[i++];

         } else {

            array[k++] = new_array_2[j++];

         }

      }

      while (i < left) {

         array[k++] = new_array_1[i++];

      }

      while (j < right) {

         array[k++] = new_array_2[j++];

      }

   }

   public static void mergeSort(int[] array, int length) {

    if (length < 2) {

         return;

      }
```

```
        int middle = length / 2;
        int[] new_array_1 = new int[middle];
        int[] new_array_2 = new int[length - middle];
        for (int i = 0; i < middle; i++) {
            new_array_1[i] = array[i];
        }
        for (int i = middle; i < length; i++) {
            new_array_2[i - middle] = array[i];
        }
        mergeSort(new_array_1, middle);
        mergeSort(new_array_2, length - middle);
        merge(array, new_array_1, new_array_2, middle, length - middle);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the size of the array: ");
        int size = sc.nextInt();
        int[] testScores = new int[size]; // Corrected the array declaration
        System.out.println("Enter the unsorted array: ");
        for (int i = 0; i < size; i++) {
            testScores[i] = sc.nextInt();
        }
        sc.close();
        System.out.println("Original Array " + Arrays.toString(testScores) + "\n");
        mergeSort(testScores, size);
        System.out.println("After Merge Sort " + Arrays.toString(testScores) + "\n");
    }
}
```

Output:

Enter the size of the array:

4

Enter the unsorted array:

44

69

22

80

Original Array [44, 69, 22, 80]

After Merge Sort [22, 44, 69, 80]

C++ Code:

```
#include <iostream>
using namespace std;
void merge(int *arr, int left, int mid, int right) {
  int temp_left[mid - left + 1];
  int temp_right[right - mid];
  for (int i = 0; i < mid - left + 1; i++) {
    temp_left[i] = arr[left + i];
  }
  for (int i = 0; i < right - mid; i++) {
    temp_right[i] = arr[mid + 1 + i];
  }
  int i = 0;
  int j = 0;
  int k = left;
  while (i < mid - left + 1 && j < right - mid) {
    if (temp_left[i] <= temp_right[j]) {
      arr[k] = temp_left[i];
      i++;
```

```
    } else {
      arr[k] = temp_right[j];
      j++;
    }
    k++;
  }
  while (i < mid - left + 1) {
    arr[k] = temp_left[i];
    i++;
    k++;
  }
  while (j < right - mid) {
    arr[k] = temp_right[j];
    j++;
    k++;
  }
}
void merge_sort(int *arr, int left, int right) {
  if (left < right) {
    int mid = (left + right) / 2;
    merge_sort(arr, left, mid);
    merge_sort(arr, mid + 1, right);
    merge(arr, left, mid, right);
  }
}
int main() {
  // Create an unsorted array.
  int arr[] = {50, 23, 12, 91, 4};
  merge_sort(arr, 0, sizeof(arr) / sizeof(arr[0]) - 1);
```

```
  for (int i = 0; i < sizeof(arr) / sizeof(arr[0]); i++) {
    cout << arr[i] << " ";
  }
  cout << endl;
  return 0;
}
```

Output:

4 12 23 50 91

Haskell Code:

```
mergelst :: [Int]->[Int]->[Int]->[Int]
mergelst [] [] z = reverse z
mergelst (x:xs) [] z = mergelst xs [] (x:z)
mergelst [] (y:ys) z = mergelst [] ys (y:z)
mergelst (x:xs) (y:ys) z = if x <= y then mergelst xs (y:ys) (x:z) else mergelst (x:xs) ys (y:z)
main = print( mergelst [ 10,30,69] [11,56,93,40] [] )
```

Output:

```
[1 of 1] Compiling Main             ( main.hs, main.o )
Linking main ...
[10,11,30,40,56,69,93]
```